

choptuik\_ac\_bc · theorems12 · [github.com/wild8highlander/choptuik\\_ac\\_bc](https://github.com/wild8highlander/choptuik_ac_bc)

# **The P4 Einstein–Hilbert Closure Theorem: the shape cycle lives inside $G = \kappa T$**

EH identities all exact · ring 4/3 and clock 1/3 derived, not postulated · frozen point not an attractor

|          |                                                                    |
|----------|--------------------------------------------------------------------|
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| Series:  | Monograph No. 19 of the theorems12<br>cycle (supplementary series) |
| Machine: | sympy_p4_einstein_hilbert.py · SymPy,<br>exact                     |
| Verdict: | einstein_direct/results/<br>p4_einstein_hilbert.json               |
| Status:  | PROVEN EXACTLY (EH layer + z-<br>coefficients) + honest audit      |

## Abstract

This supplementary monograph records the P4 Einstein–Hilbert closure theorem: the shape cycle of the corrected tower lives inside the single Einstein system  $G = \kappa T$ . The EH layer verifies exactly:  $T_{uv} = 0$  identically (the null-null sector of the Hilbert tensor is empty), the Hilbert identities  $\square\Phi-v = (\kappa/2)T-v$  hold for  $v = u, v$ , box $\Phi - SC$  closes, and each station equation is a component of one system:  $G_{uv} \propto UV + rTH$ ,  $G_{uu} - \kappa T_{uu} \propto C1$ ,  $G_{vv} - \kappa T_{vv} \propto C2$ , all with the factor  $-2/r(u,v)$ .

The z-coefficient status of all twelve chain equations is ok (no silently zeroed derivatives in the corrected system), and the honest audit is recorded: the ring  $4/3 = W2h/T0h^2/R1h$  and the clock  $P2h/T0h = 1/3$  are exact consequences of the O1–O5 core — the machine derives them, not postulates; the session-11 book construction ( $\kappa_{cyc} = \ln(64/9)$ ) on the branches  $9/4$  and  $9/16$ ) relied on an artifact — the branches do not exist separately in the corrected system; the spectrum shows  $\lambda^+(\tau)$  monotone with  $\lambda^+ = \kappa_{obs}$  at  $\tau \approx 0.486$  and  $\lambda^+ \approx 8-9$  at  $\tau^* = 27/4$  — the frozen point is not a critical attractor by spectrum. Tables, corollaries, honest caveats and the reproduction protocol are included.

**Keywords:** Einstein–Hilbert closure, shape cycle,  $G = \kappa T$ , torner ring  $4/3$ , session-11 audit, spectrum  $\lambda^+(\tau)$ , SymPy, machine theorem.

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## 1. Introduction

This supplementary monograph records the P4 Einstein–Hilbert closure theorem — the machine result of the P4 campaign (machine `sympy_p4_einstein_hilbert.py`, verdict `p4_einstein_hilbert.json`, LEDGER §20). The P4 closure (order O6+) is the shape cycle of the tower: the question was how the cycle lives in the equations — as an external ansatz imposed on  $G = \kappa T$ , or as modes of the system itself.

The machine answer: every station of the cycle is a mode of ONE Einstein system — the dictionary of operators ( $d_u/d_v$ ,  $c/d$ ,  $s/t$ ,  $p/q$ ) maps the station equations to components of  $G = \kappa T$  with exact factors  $-2/r$ . This upgrades the cycle from an ansatz to a derived structure, and it exposes the two honesty points the audit records: the ring and the clock are derived (not postulated), and the session-11 two-branch book construction was an artifact — the corrected system does not split into separately existing branches.

## 2. Notation and Setting

The P4 model is the adiabatic shape cycle of the O6+ tower: the second flows  $ddW2/ddR3/ddP4 = 0$  (the tables  $(5dF + 6F)/S^4$ ), in the variants V1 (CSS frame + live sources) and V2 (full dynamics, all flows live). The EH layer checks the identities of the Hilbert derivation against the chain:  $T_{uv} = 0$ , the  $\square\Phi$ -identities, and the proportionality of each station equation to a component of  $G - \kappa T$ .

The convention  $\kappa = 2$  is the Burko normalization of the numerical code ( $\kappa = 8\pi G$  in physics units); the factor  $-2/r(u,v)$  is the universal proportionality of the station equations to the Einstein components.

Table 19.1. The Einstein–Hilbert checks (`einstein_hilbert.checks`)

| Check                                | Value | Status |
|--------------------------------------|-------|--------|
| <code>T_uv_is_zero</code>            | true  | exact  |
| <code>hilbert_identity_nu = u</code> | true  | exact  |
| <code>hilbert_identity_nu = v</code> | true  | exact  |
| <code>boxPhi_minus_SC</code>         | true  | exact  |

| Check                             | Value                 | Status |
|-----------------------------------|-----------------------|--------|
| G_uv_proportional_UV_plus_r<br>TH | true (factor $-2/r$ ) | exact  |
| G_uu_minus_kT_proportional_<br>C1 | true (factor $-2/r$ ) | exact  |
| G_vv_minus_kT_proportional_<br>C2 | true (factor $-2/r$ ) | exact  |

### 3. Statement of the Theorem

Theorem 19 (the P4 Einstein–Hilbert closure). For the corrected tower in the P4 model:

- (i) the stations are modes of one system: each of the twelve chain equations is proportional (factor  $-2/r(u,v)$ ) to a component of  $G = \kappa T$  — the amplitude book (C1/C2, sectors uu/vv) and the UV/TH channel (uv) hold on different components of one Hilbert tensor;
- (ii) the null-null sector is empty identically:  $T_{uv} = 0$ , so the uv-channel is carried by the geometric side ( $UV + rTH$ ) alone; the Hilbert identities  $\square\Phi\text{-}v = (\kappa/2)T\text{-}v$  hold exactly for  $v = u, v$ ;
- (iii) the ring and the clock are derived:  $4/3 = W2h/T0h^2/R1h$  and  $P2h/T0h = 1/3$  follow exactly from the O1–O5 core ( $Mdef\_ξ3 \rightarrow D0 = 0$ ;  $UV\_ξ1$  &  $C2\_ξ1$ ;  $C2\_ξ2$ ) — the machine derives them, it does not postulate them;
- (iv) the z-coefficient status of all twelve chain equations is ok in the corrected system — the session-8 audit (second derivatives silently zeroed by the TBL\_F substitution) is fixed, and no equation loses its z-content;
- (v) the honest audit stands: the session-11 book construction  $\kappa_{cyc} = \ln(\tau^3/\tau^5) = \ln(64/9)$  on the branches 9/4 and 9/16 relied on an artifact — the branches do not exist separately in the corrected system; and the spectrum  $\lambda^+(\tau)$  is monotone with  $\lambda^+ = \kappa_{obs} (\approx 1.96)$  at  $\tau \approx 0.486$  and  $\lambda^+ \approx 8-9$  at  $\tau^* = 27/4$  — the frozen point is not a critical attractor by spectrum; the O7 prolongation freezes the adiabatic dynamic points ( $\sigma_{W2} - 2\sigma_{T0} - \sigma_{R1} = -1.53$  in V2).

## 4. Sketch of the Machine Proof

The machine builds the Einstein–Hilbert components symbolically from the metric  $ds^2 = -\alpha^2 du dv + r^2 d\Omega^2$  with the Hilbert stress-energy, subtracts  $\kappa T$  from each G-component, substitutes the chain amplitudes, and factors: each difference collapses to  $-2/r$  times the corresponding station equation. The z-coefficient layer then checks that every equation retains its z-content (the corrected TBL\_F substitution), and the spectrum layer evaluates  $\lambda^+(\tau)$  of the frozen pencil. The audit layer records the session-8 fix and the session-11 artifact verdicts verbatim.

Table 19.2. The audit record (honest\_notes digest)

| Audit item                                    | Verdict                                                                                                                                     |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Session 8: second derivatives                 | W2''/R3''/P4'' and R5' were silently zeroed by TBL_F; fixed — all 12 z-statuses ok                                                          |
| Session 11: $\kappa_{\text{cyc}} = \ln(64/9)$ | relied on separate branches 9/4, 9/16 — artifact; branches do not exist separately                                                          |
| Ring 4/3 and clock 1/3                        | exact consequences of the O1–O5 core (derived, not postulated)                                                                              |
| Spectrum $\lambda^+(\tau)$                    | monotone; $\lambda^+ = \kappa_{\text{obs}}$ at $\tau \approx 0.486$ ; $\lambda^+ \approx 8\text{--}9$ at $\tau^* = 27/4$ — not an attractor |
| O7 prolongation                               | freezes adiabatic dynamic points ( $\sigma_{\text{W2}} - 2\sigma_{\text{T0}} - \sigma_{\text{R1}} = -1.53$ in V2)                           |

## 5. Numerical Results

The EH layer is exact (no numerics beyond the symbolic simplification); its seven checks of Table 19.1 all return true. The z-coefficient status table lists ok for all twelve equations: SC\_ξ4, UV\_ξ1, UV\_ξ3, C2\_ξ1, C2\_ξ2, C2\_ξ3, TH\_ξ2, C1\_ξ1, C1\_ξ2, C1\_ξ3, Mdef\_ξ3, Mdef\_ξ5. The anchors layer confirms the machine context: o6nsolve, hexcycle, deriv, hexcheck, hex and hex\_modes all present.

The spectrum numbers frame the dynamics honestly: at the frozen point  $\tau^* = 27/4$  the growing branch reaches  $\lambda^+ \approx 8\text{--}9$  — an extrapolation outside the computed grid — while

at  $\tau \approx 0.486$  it crosses  $\kappa_{\text{obs}} \approx 1.96$ ; between the two the tower is spectrally alive but dynamically frozen by the O7 conservation law. The frozen point is thus an algebraic vertex, not an attractor — consistent with the holonomy theorem (No. 7).

## 6. Corollaries

Corollary 19.1 (the cycle is not an ansatz). Since every station is a component of  $G = \kappa T$ , the shape cycle inherits the Bianchi consistency of the Einstein system: any future station set must map back into the same dictionary, or it is a different theory.

Corollary 19.2 (the audit discipline is part of the theorem). The session-8 fix and the session-11 retraction are recorded inside the verdict: the corrected system's z-content is certified, and the two-branch book construction is closed as an artifact — the honest ledger is machine-carried, not editorial.

Corollary 19.3 (no spectral escape at the frozen point). The monotone  $\lambda_+(\tau)$  with  $\lambda_+(\tau^*) \approx 8\text{--}9$  closes the route to an exponential repeller at  $\tau^*$ : any escape must be finite-amplitude — the standing separation of the whole series.

## 7. Honest Caveats

The theorem is proven for the P4 model (the adiabatic second flows  $ddW2/ddR3/ddP4 = 0$ ) in the two variants V1/V2; a non-adiabatic P4 prolongation is the O7 programme and is outside this verdict. The  $\lambda_+$  values at  $\tau^*$  are extrapolations beyond the computed grid (honest note of the verdict). The session-11 retraction covers the book construction on the branches; other uses of  $\kappa_{\text{cyc}}$  (the figure-cycle anchors of monograph No. 14) are not affected. The EH layer uses the Burko normalization  $\kappa = 2$ ; translating to  $\kappa = 8\pi G$  is a units statement. Finally, the audit language (“artifact”, “derived”) is the machine's own honesty vocabulary, quoted verbatim.

## 8. Reproducibility

Table 19.3. Reproduction protocol for Theorem 19

| Step | Command                                 | Result                                                           |
|------|-----------------------------------------|------------------------------------------------------------------|
| 1    | python3<br>sympy_p4_einstein_hilbert.py | results/p4_einstein_hilbert.json<br>(EH layer + audit)           |
| 2    | python3 sympy_derivation.py             | results/derivation_results.json<br>(the Hilbert identity source) |
| 3    | python3 -m pytest tests/ -q             | 58 passed (P4 tests included)                                    |

The machine runs in minutes (deterministic, offline). The data correspond to commit 5b256cd; an autonomous copy lives in `einstein_direct/open9/machines/`.

## References

- [1]                      Repository                      wild8highlander/choptuik\_ac\_bc:  
einstein\_direct/sympy\_p4\_einstein\_hilbert.py;  
einstein\_direct/results/p4\_einstein\_hilbert.json (LEDGER §20).
- [2]                      Ibid.:                      einstein\_direct/sympy\_derivation.py;  
einstein\_direct/results/derivation\_results.json (the Hilbert identity).
- [3] Monographs No. 7, No. 14, No. 18 of the theorems12 cycle.